

Probability: Worked Solutions

Problem 1. Moments and conditioning from a joint law

Problem. Let $X, Y \in \{0, 1\}$ have joint probabilities

	$Y = 0$	$Y = 1$	p_X
$X = 0$	0.42	0.18	0.60
$X = 1$	0.08	0.32	0.40
p_Y	0.50	0.50	1

Calculate $\mathbb{E}X$, $\mathbb{E}Y$, $\mathbb{E}[XY]$, $\text{Cov}(X, Y)$, and $\mathbb{P}(X = 0 \mid Y = 1)$. Are the variables independent?

Solution. Because X and Y are binary, their expectations equal their success probabilities:

$$\mathbb{E}X = 0(0.60) + 1(0.40) = 0.40, \quad \mathbb{E}Y = 0(0.50) + 1(0.50) = 0.50.$$

The product XY is one only in the cell $(1, 1)$, so $\mathbb{E}[XY] = 0.32$. Therefore

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}X\mathbb{E}Y = 0.32 - (0.40)(0.50) = 0.12.$$

The event $Y = 1$ has probability 0.50. Restricting to that column and normalizing gives

$$\mathbb{P}(X = 0 \mid Y = 1) = \frac{0.18}{0.50} = 0.36.$$

Independence would require every joint cell to be the product of its marginals. Here $\mathbb{P}(X = 1, Y = 1) = 0.32 \neq 0.40 \cdot 0.50 = 0.20$, so the variables are dependent. Their nonzero covariance also rules out independence.

Problem 2. A conditional Gaussian law

Problem. Let Z_1, Z_2 be independent standard Gaussian variables. Set $X = 2Z_1$ and $Y = Z_1 + Z_2$. Find the covariance matrix of $(X, Y)^T$ and the conditional law of X given $Y = y$.

Solution. The means are zero. Independence gives $\text{Cov}(Z_1, Z_2) = 0$, hence

$$\text{Var}(X) = 4, \quad \text{Var}(Y) = 1 + 1 = 2, \quad \text{Cov}(X, Y) = \text{Cov}(2Z_1, Z_1 + Z_2) = 2.$$

Since a linear transformation of a Gaussian vector is Gaussian,

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}\right).$$

A direct decomposition gives the conditional law. Put $R = X - Y = Z_1 - Z_2$. Then $\text{Var}(R) = 2$ and

$$\text{Cov}(R, Y) = \text{Cov}(Z_1 - Z_2, Z_1 + Z_2) = 1 - 1 = 0.$$

The pair (R, Y) is jointly Gaussian, so zero covariance implies independence. Thus $X = Y + R$, with $R \sim \mathcal{N}(0, 2)$ independent of Y , and

$$\boxed{X \mid Y = y \sim \mathcal{N}(y, 2).}$$

Equivalently, the conditional mean is $(2/2)y = y$ and the conditional variance is the Schur complement $4 - 2(1/2)2 = 2$.

Problem 3. Dependence through a shared parameter

Problem. Let P be equally likely to be $1/4$ or $3/4$. Conditional on P , draw X_1, X_2 independently, each from $\text{Bernoulli}(P)$. Compute $\mathbb{P}(X_1 = X_2 = 1)$ and $\text{Cov}(X_1, X_2)$.

Solution. Conditional independence gives $\mathbb{P}(X_1 = X_2 = 1 \mid P) = P^2$. Averaging over the two possible parameter values,

$$\mathbb{P}(X_1 = X_2 = 1) = \mathbb{E}[P^2] = \frac{1}{2} \left(\frac{1}{16} + \frac{9}{16} \right) = \frac{5}{16}.$$

By total expectation, $\mathbb{E}X_i = \mathbb{E}[\mathbb{E}(X_i \mid P)] = \mathbb{E}P = 1/2$. Since X_1X_2 is the indicator of two successes,

$$\text{Cov}(X_1, X_2) = \frac{5}{16} - \frac{1}{2} \cdot \frac{1}{2} = \boxed{\frac{1}{16}}.$$

This equals $\text{Var}(P)$. Observing one success makes the larger shared probability more plausible, increasing the marginal probability of another success. Consequently X_1 and X_2 are dependent even though they are independent after conditioning on P .

Problem 4. Transforming a uniform variable

Problem. For $U \sim \text{Unif}(0, 1)$, derive the density of $Y = U^2$ and verify that it integrates to one. Compute its expectation using both the new density and the law of the unconscious statistician (LOTUS).

Solution. On $(0, 1)$ the transformation $u \mapsto u^2$ is increasing, with inverse $u = \sqrt{y}$. Thus, for $0 < y < 1$,

$$F_Y(y) = \mathbb{P}(U^2 \leq y) = \mathbb{P}(U \leq \sqrt{y}) = \sqrt{y}.$$

The CDF is zero for $y \leq 0$ and one for $y \geq 1$. Differentiating on $(0, 1)$ gives

$$f_Y(y) = \frac{1}{2\sqrt{y}} \mathbf{1}\{0 < y < 1\}.$$

The density diverges near zero, but the improper integral is finite:

$$\int_0^1 \frac{dy}{2\sqrt{y}} = [\sqrt{y}]_0^1 = 1.$$

Its endpoint values do not affect the distribution. Integrating against this density,

$$\mathbb{E}Y = \int_0^1 y \frac{dy}{2\sqrt{y}} = \frac{1}{2} \int_0^1 y^{1/2} dy = \frac{1}{3}.$$

LOTUS calculates the same expectation directly under the original law:

$$\mathbb{E}[U^2] = \int_0^1 u^2 du = \frac{1}{3}.$$

There is only one inverse branch here because $U > 0$; a variable supported on both sides of zero would require both square roots.

Problem 5. Sample size and Monte Carlo precision

Problem. An iid Monte Carlo estimate has standard error 0.02 at $n = 2500$. Estimate the sample size required for standard error 0.005 with unchanged integrand variance. Explain the different roles of sample size and simulation repetitions.

Solution. Write the estimate as $\hat{\mu}_n = n^{-1} \sum_{i=1}^n g(X_i)$, where the draws are independent and $\text{Var}(g(X_i)) = \tau^2 < \infty$. Then

$$\text{se}(\hat{\mu}_n) = \frac{\tau}{\sqrt{n}}, \quad \frac{0.005}{0.02} = \sqrt{\frac{2500}{n_{\text{new}}}}.$$

Reducing standard error by a factor of four requires sixteen times as many draws:

$$n_{\text{new}} = 2500 \left(\frac{0.02}{0.005} \right)^2 = 40000.$$

The given numbers imply $\tau \approx 0.02\sqrt{2500} = 1$; if the original standard error was estimated, 40000 is correspondingly a planning estimate.

Now repeat an entire n -draw computation B times, obtaining B independent estimates. Their spread estimates the standard error of one n -draw estimate. Increasing B improves the precision with which this spread or the sampling distribution is measured; it leaves the variance τ^2/n of each constituent estimate unchanged. If the B estimates are themselves averaged, that is a new estimator using Bn independent draws and has variance $\tau^2/(Bn)$.